AURATHON 2025

Team Name : NewDev
Members :
-Naina Mansukhani
-Harneet Kaur
-Piyush Singh
-Yashu
College : Guru Tegh Bahadur Institute Of Technology
Contact Details :
Email id- nainamansukhani123@gmail.com
Phone number- 9315885986



PROBLEM STATEMENT



•Global crop losses reach 30-40% annually due to undetected plant diseases

•Farmers lose $2.5 trillion worldwide each year from preventable crop damage

•Traditional forecasting models rely on historical averages, ignoring real-time conditions

•Static predictions show 15-25% error rates, leading to poor resource planning

•Sudden weather changes, pest outbreaks, and disease pressure are not factored in

Technology Access Gap:

•Advanced agricultural technologies cost $5,000-50,000 per farm to implement

•Complex systems require technical expertise that rural farmers often lack

PROPOSED SOLUTION

1.Environmental Context Embeddings

. Proprietary algorithms convert environment data into disease risk indicators

•Creates unique "environmental fingerprints" for pathogen-favorable conditions

•Transforms traditional reactive detection into proactive prediction

2. Context-Aware Computer Vision

•Novel fusion technology combining leaf image analysis with environmental data

•Predicts disease susceptibility before visual symptoms appear

3. Predictive Risk Engine

•48-hour advance disease outbreak warnings with probability scores

•Real-time risk heat maps showing exactly where problems will emerge

•Dynamic yield forecasts updated based on predicted disease impact

OUR IDEA IN ACTION

Environmental Context Embeddings (Our Core AI Innovation)

- Neural Network Encoders convert soil moisture, temperature, humidity data into numerical vectors
- Deep Learning Models identify disease favourable environmental patterns from historical data
- Predictive Algorithms forecast optimal pathogen development conditions 48 hours ahead

Multi-Modal Fusion AI System

- Computer Vision Models extract visual features from crop leaf images using CNNs
- Environmental Feature Networks process real-time sensor data through custom neural architectures
- Attention Mechanisms determine which environmental factors most influence disease risk
- Fusion Layers combine visual and environmental data for enhanced prediction accuracy
- Real-Time Learning continuously improves predictions based on new field data



FEASIBILITY AND VIABILITY

Technical feasibility

•Mobile-Ready: Optimized for smartphone hardware with offline processing capability

•Scalable Infrastructure: Cloud-based system supporting 1-10,000-hectare operations

•Low Implementation Cost: $50-100 per farm using existing smartphone infrastructure

Business Potential:

•Scalability: Single platform works across all crop types and geographical regions

•Partnership Opportunities: Integration with equipment manufacturers, seed companies, insurance providers

•Government Support: Agricultural digitization initiatives in developing countries provide funding opportunities

Economic Viability:

•Proven AI Foundation: Built on established CNN/Transformer architectures

•Mobile-Ready: Optimized for smartphone hardware with offline processing capability

•Scalable Infrastructure: Cloud-based system supporting 1-10,000 hectare operations